



Published in final edited form as:

Aphasiology. 2022 ; 36(11): 1351–1372. doi:10.1080/02687038.2021.1966374.

Factors predicting long-term recovery from post-stroke aphasia

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Abstract

Background: It remains widely accepted that spontaneous recovery from aphasia is largely limited to the first related factors. This has direct implications for acute and chronic interventions for aphasia. few months following stroke. A few recent studies challenge this view, revealing that some individuals' language abilities improve even during the chronic stage.

Aims: To identify prognostic indicators of long-term aphasia recovery.

Methods & Procedures: Eighteen people with aphasia initially evaluated in the chronic stage were retested at least one year later. The Western Aphasia Battery-Revised (WAB-R) Aphasia Quotient (AQ) was used to quantify changes in language impairment. Prognostic factors included those related to the patient (demographic, psychosocial), stroke (lesion volume and location), and treatment (medical, rehabilitative).

Outcomes & Results: Twelve participants improved and 6 remained stable or declined. Linear regression analysis revealed that lesion volume predicted long-term language gains, with smaller lesions yielding greater improvements. Individuals who did not improve were more likely to have lesions encompassing critical frontal and temporoparietal cortical regions and interconnecting white matter pathways. Exploratory regression analysis of psychosocial and treatment-related factors revealed a positive relationship between improvement and satisfaction with life participation, and a negative relationship between improvement and perceived impairment severity. Critically, psychosocial and treatment-related factors significantly improved model fit over lesion volume, suggesting that these factors add predictive value to determining long-term aphasia prognosis.

Conclusions: Long-term aphasia recovery is multidetermined by a combination of stroke-, psychosocial-, and treatment-related factors. This has direct implications for acute and chronic interventions for aphasia.

Keywords

Aphasia; stroke; neuroplasticity; neurological rehabilitation; acquired language disorder; long-term recovery

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Disclosure statement

The authors report no conflict of interest.

Introduction

Chronic stroke-induced aphasia affects approximately one million individuals in the US alone. It is generally accepted that most language improvements occur within the first three months post-stroke onset (MPO), due largely to spontaneous recovery (El Hachoui et al., 2013; Laska et al., 2001; Lazar et al., 2008, 2010; Pedersen et al., 1995, 2004; Teasell et al., 2012; also see Kiran & Thompson, 2019 for a review). Thereafter, gains are presumed to be smaller and are primarily associated with formal speech-language therapy (SLT; Allen et al., 2012; Brady et al., 2016; Breitenstein et al., 2017; Robey, 1994, 1998; Teasell et al., 2012). This view of chronic aphasia as static directly impacts the availability of therapy and other supports for patients with persistent language deficits (Code & Petheram, 2011; Richard et al., 2000). Recently, however, we discovered that some individuals with chronic aphasia made substantial recovery that cannot, to the best of our knowledge, be ascribed to formal therapy alone. While there have been a small number of studies in the past few years to report similar findings of aphasia recovery in the chronic stage (Basilakos et al., 2019; Holland et al., 2017; Hope et al., 2017; Johnson et al., 2019), the prognostic indicators of long-term recovery remain to be clarified. Identifying factors that predict the potential for aphasia recovery in the long term has clear clinical implications, as it can inform both acute and chronic rehabilitation approaches. The purpose of this study was to add to the small, but growing, body of literature investigating long-term aphasia recovery, and explore potential prognostic indicators of long-term recovery in chronic aphasia.

Most evidence regarding aphasia recovery comes from studies that have focused on the first year post-stroke onset (Lazar et al., 2008, 2010; Pedersen et al., 1995, 2004; Wade et al., 1986). Approximately one third of stroke survivors present with aphasia acutely (Berthier, 2005), and nearly half present with language impairments that persist beyond the first three MPO. Reported language gains after this time period are much smaller, and are thought to plateau by six MPO (Laska et al., 2001; Maas et al., 2012; Pedersen et al., 1995) – especially in the absence of continuing formal SLT (Robey, 1994, 1998). Consequently, the prevailing view is that spontaneous recovery from aphasia is limited to the first few months after stroke onset.

Recently, however, this view of the time course of aphasia recovery has been challenged (Basilakos et al., 2019; Holland et al., 2017; Hope et al., 2017; Johnson et al., 2019). Holland et al. (2017) discovered that some individuals' language performance improved during the chronic stage, as assessed by the Western Aphasia Battery-Revised (WAB-R; Kertesz, 2006) and other measures of discourse. They studied 26 individuals with chronic aphasia who attended aphasia centers affiliated with the AphasiaBank project (MacWhinney et al., 2011), and found that 16 individuals' language abilities improved, while 10 individuals either remained stable ($n = 7$) or declined ($n = 3$). The finding that aphasia severity can improve in the chronic stages has since been reported in at least three other studies (Basilakos et al., 2019; Hope et al., 2017; Johnson et al., 2019). Thus, increasing evidence points to the potential for continued language recovery in chronic aphasia in the absence of extended formal SLT, indicating that the commonly accepted notion that aphasia stabilises in the chronic phase may be the result of prior studies not following chronic patients over sufficient periods of time to detect performance changes.

Aphasia recovery in the chronic stage is highly variable (Kiran & Thompson, 2019), underscoring the importance of identifying factors associated with long-term outcomes. Prior work investigating prognostic variables in the first year post stroke has focused on factors related to the patient, such as age, sex, and education as well as those related to the stroke, i.e. lesion volume and location. However, the prognostic value of these factors is debated (Plowman et al., 2012). For instance, some studies indicate a relationship between age and aphasia prognosis (Laska et al., 2001, 2008; Pedersen et al., 1995), whereas others do not (Lazar et al., 2008, 2010; Lendrem & Lincoln, 1985). Other patient-related factors, such as sex and education, do not appear to significantly contribute to aphasia recovery (Lazar et al., 2008; Lendrem & Lincoln, 1985; Pedersen et al., 1995, 2004), but have been associated with initial aphasia severity (Plowman et al., 2012). Similarly, the evidence regarding the impact of stroke-related variables on recovery is also mixed. Previous studies indicating that lesion volume does not account for aphasia recovery assessed the predictive value of this factor using an approximate measure of lesion volume (Laska et al., 2001; Lazar et al., 2008, 2010), which may have affected the reliability of their results. In a comprehensive review of the aphasia recovery literature, Plowman et al. (2012) suggest that lesion site and size are the most consistent predictors of language outcomes, citing large-scale studies that demonstrate a strong relationship between lesion volume and eventual aphasia severity.

Whether these patient- and stroke-related variables constitute prognostic indicators of aphasia recovery in the long term has yet to be determined. Indeed, the few studies to report cases of aphasia recovery in the chronic stage have investigated some of the aforementioned prognostic factors; however, here too, the results have been mixed. For instance, some evidence suggests that age is inversely related to long-term outcomes – though these studies differ in whether the significant relationship was with age at stroke (Johnson et al., 2019) or age at the time of assessment (Basilakos et al., 2019), yet other studies do *not* find age to be a predictor of long-term change in aphasia severity (Holland et al., 2017; Hope et al., 2017). However, other patient-related variables, such as sex (Hope et al., 2017) and education (Basilakos et al., 2019; Johnson et al., 2019), do not appear to influence recovery in the chronic stage, which is largely consistent with earlier work that followed individuals with aphasia over shorter time frames. Regarding stroke-related factors, the two studies to explore lesion volume found no relationship between this factor and long-term recovery patterns (Basilakos et al., 2019; Hope et al., 2017). To our knowledge, no study has examined the impact of lesion location on long-term aphasia recovery in the same people with aphasia assessed at different time points in the chronic stage (see Hillis et al., 2018 for evidence from a cross-sectional study). Given this incomplete and somewhat conflicting account of the prognostic indicators of chronic aphasia recovery, further investigation of previously identified patient- and stroke-related factors is needed.

Evidence shows that chronic aphasia increases the need for family assistance in daily activities and decreases overall perceptions of quality of life (Cruice et al., 2003; Tennant et al., 1997), which may also represent a patient-related variable critical in mediating some aspects of chronic aphasia recovery. Holland et al. (2017) speculated that life participation and other psychosocial factors might have played an important role in the long-term aphasia recovery patterns observed in their sample of people with aphasia. Indeed, communication

abilities in individuals with chronic aphasia influence perceived quality of life due in part to the limitations aphasia places on life participation (Cruice et al., 2003). Community programs and aphasia centers provide individuals with aphasia the opportunity to participate in activities and engage with one another, which may serve to enhance recovery when formal SLT is no longer available due to lack of benefits or finances. Aphasia centers have been shown to have a positive impact on both quality of life and communication abilities (Van Der Gaag et al., 2005). Thus, it is important to establish whether patient-related factors relating to psychosocial variables are associated with aphasia recovery in the long term.

The goal of the current study was to retrospectively explore factors associated with changes in aphasia severity following our incidental finding that some individuals' language abilities substantially improve during the chronic stage. The participants in this study were part of a larger cohort of volunteers who had earlier participated in a research study conducted at Moss Rehabilitation Research Institute and the University of Pennsylvania and given permission to be contacted again for future studies. In the course of retesting persons in this cohort to determine their eligibility for a new study, we noted that some exhibited marked improvement on the WAB-R (Kertesz, 2006) despite having been considered "chronic" at the initial administration. This unexpected finding provided us with the opportunity to explore factors that may promote long-term spontaneous recovery from aphasia. Following prior work (Basilakos et al., 2019; Holland et al., 2017; Johnson et al., 2019), we used the WAB-R Aphasia Quotient (AQ) to quantify changes in aphasia severity from initial to follow-up language assessment. Demographic and structural neuroimaging data collected as part of the original protocol allowed us to examine the influence of previously identified patient-related (age, sex, education) and stroke-related factors (lesion volume and location) on aphasia recovery in the chronic phase. We also assessed whether time since stroke onset (i.e., MPO) at initial testing was related to later recovery, as people with aphasia initially evaluated in close temporal proximity to stroke onset may show larger language gains due to the peak in neuroplastic mechanisms that occurs during the first few months following stroke (Kiran & Thompson, 2019). To explore whether patient-related variables reflecting psychosocial factors and/or prior stroke-related treatment mediate long-term recovery, we added self-report measures of quality of life and treatment history to the testing protocol, and continued with the follow-up evaluation of all those who qualified and remained available for re-test. This study reports results from this convenience sample.

Materials and methods

Participants

Participants had previously participated in a research study, which included administration of the WAB-R and a research quality MRI or CT scan. As participants in the Neuro-Cognitive Rehabilitation Research Patient Registry at the Moss Rehabilitation Research Institute (Schwartz et al., 2005), they had consented to have their demographic and research data stored and reviewed for the purpose of determining eligibility for future research projects. All had been diagnosed acutely with aphasia due to left hemisphere stroke and suffered from language deficits at the time they were enrolled in the original study. Twenty-

three individuals from the above registry gave informed consent to participate in a multi-session protocol approved by the IRB at Albert Einstein Medical Center.

Participants were included based on the following criteria: initial language assessment obtained at 5 or more MPO; native English speaker, pre-morbid right handedness, normal or corrected-to-normal vision and hearing; no other comorbid neurological or psychiatric disorders; at least 1 year time interval between initial and follow-up testing; and had undergone CT or MRI scanning at the time of the initial language assessment. Five participants were excluded from the study; three did not meet the above criteria; one moved before completing testing; and one reported ongoing life stressors that affected language performance and responses on psychosocial measures. The 18 individuals included in the study (12 male) had a mean age of 52.9 (standard deviation (SD) = ± 12.7 ; range = 30–77), years of education of 15 (SD = ± 3.2 ; range = 12–21), and MPO of 39.8 (SD = ± 39.2 ; range = 5–133) at the initial assessment. See Table 1 for demographic and clinical information.

Behavioral Measures

A certified speech-language pathologist or trained research assistant administered and scored the WAB-R and self-report measures of psychosocial variables and treatment history.

At the follow-up assessment, psychosocial variables were evaluated with the Assessment for Living with Aphasia (ALA), a standardised pictographic questionnaire from the Aphasia Institute in Toronto that rates the self-perceived burden of aphasia as it pertains to quality of life and life participation (Kagan et al., 2011). The ALA queries people with aphasia using a pictographic aid in order to obtain a self-report measure of the experience of living with aphasia (Simmons-Mackie et al., 2014). For all questions, participants gave their responses on a 4-point scale where 0 reflects a negative response (i.e., “Never”, “Big Problem”, “Definitely No”, “Not Happy”, or “Very Bad”) and 4 reflects a positive response (i.e., “Always”, “No Problem”, “Definitely Yes”, “Very Happy”, or “Very Good”). The ALA gives a computed total score that represents a composite score of different psychosocial domains reflecting the following constructs: communication and language environment (i.e., environment domain), participation in life situations (i.e., participation domain), personal identity, attitudes and feelings (i.e., personal domain), language and related impairments (i.e., aphasia domain), and a “moving on with life” question, which represents the overall impact of aphasia on quality of life using a metaphor of traversing a brick wall.

We created “add-on” questions, adopting the ALA pictographic format to gain insight from participants concerning other factors that may impact recovery, including prior stroke-related treatments and participation in other activities not queried by the ALA. A subset of the add-on items used a multiple choice response format (“yes”, “no”, and “not sure”) to query if participants received the “clot-busting” drug (i.e., tissue plasminogen activator [tPA]) upon hospital admission; attended rehabilitation upon hospital release; received outpatient SLT; attended a residential program for their aphasia; and participated in aphasia center activities. The remaining add-on questions used the ALA scale to probe how often people with aphasia participated in research, their satisfaction with learning about/using technology, the frequency with which they use technology to practice their speech, and how many courses of

SLT they received. The add-on questions were administered during the same session as the ALA.

Neuroimaging Data

Structural imaging data for all participants were obtained within 6 months of initial testing as part of the original study protocol. We collected high-resolution whole-brain T1-weighted images obtained on a 3T Siemens Trio scanner from 15 participants. CT scans were obtained for the remaining 3 participants who were MRI ineligible. Lesions obtained via MRI were manually segmented on the structural image by a trained technician, and checked for accuracy by an experienced neurologist blinded to the behavioral data. The same neurologist manually traced lesions obtained via CT. Lesion tracings were then warped to a standard atlas space (i.e., the Montreal Neurological Institute (MNI) Colin27 template) using a symmetric diffeomorphic registration algorithm (<http://www.picssl.upenn.edu/ANTS/>; Avants et al., 2006). The VoxBo brain imaging package (<https://github.com/kimberg/voxbo>) was used to compute lesion volume.

Statistical analyses

Regression analyses—We conducted three multiple linear regression analyses in order to (1) assess the predictive value of previously identified patient-related demographic factors (age, sex, education) and stroke-related factors (lesion volume) on long-term outcomes while accounting for MPO; (2) explore the relationship between aphasia recovery and patient-related psychosocial factors (ALA responses, research and aphasia center participation, technology use) as well as treatment-related variables (amount of SLT received, technology use to practice speech); and (3) evaluate whether psychosocial and treatment-related variables add predictive value to long-term recovery above and beyond the previously identified factors found to significantly predict recovery. The rationale for excluding some of the aforementioned psychosocial and treatment-related factors is described in more detail below. In both regression models, aphasia recovery was defined as percent change in WAB-R AQ from initial to follow-up language assessment, calculated as $((AQ_{\text{Follow-up}} - AQ_{\text{initial}}) / AQ_{\text{initial}}) \times 100$.

To evaluate whether previously identified patient- and stroke-related factors predict long-term outcomes, we entered the following predictor variables: lesion volume, MPO at initial assessment, age at stroke, years of education, and sex. Age at initial assessment was not included because (1) this factor was highly correlated with stroke age ($p < .001$), and (2) there is reason to believe that age at stroke would more strongly influence the potential for compensatory neuroplastic responses, and therefore be a better predictor of long-term language gains when compared to age at the initial assessment.¹

Due to the exploratory nature of the regression analysis performed on patient-related psychosocial and treatment-related variables, we used backward stepwise regression, which included the following variables: responses on the ALA subdomains (aphasia, participation,

¹Age and MPO at the follow-up assessment were not included in the analysis, as these factors are perfectly correlated with age and MPO at the initial assessment, respectively.

environment, and personal); responses to the add-on questions probing aphasia center participation, prior inpatient SLT, frequency of research participation, satisfaction with learning about/using technology generally or specifically to practice speech, and amount of SLT received. The ALA total score was not included because this value is calculated from the subdomain scores, and thus represents a non-independent predictor. Similarly, the single “moving on with life” question was excluded from the regression analysis because scores are based on a single question, and the interpretation of responses on this question is not entirely clear (Simmons-Mackie et al., 2014). Lastly, the multiple-choice “add-on” questions except for aphasia center participation and prior inpatient SLT were not included in the model due to either near zero variance in responses or too many missing data points (i.e., “not sure” responses; see Supplementary Table 1 for a breakdown of responses).

The third regression model aimed to assess the predictive value of psychosocial and treatment-related factors when compared to patient- and stroke-related factors previously identified as impacting recovery. For this analysis, we used a stepwise regression where we first entered factors that significantly contributed to aphasia recovery in the model assessing previously identified factors impacting recovery followed by the addition of significant predictors in the exploratory regression analysis of psychosocial and treatment-related factors. This allowed us to evaluate whether psychosocial and/or treatment-related factors improve model fit over known patient- and/or stroke-related factors previously identified in the literature without compromising statistical power to detect an effect of any one factor.

For all regression models, the assumption of normally distributed residuals was assessed using histograms and normal probability plots (i.e., P-P plots). Variables were removed from the model if their inclusion resulted in non-normally distributed residuals. The multiple linear regression analyses were performed using IBM SPSS Statistics for Macintosh, version 24 (IBM Corp., Armonk, N.Y., USA).

Lesion location analyses—To determine if lesion location is associated with the potential for long-term language gains, we adopted two lesion symptom mapping approaches: lesion subtraction analysis (Karnath et al., 2002) and voxel-based lesion-symptom mapping (VLSM; e.g., Bates et al., 2003; Dronkers et al., 2004). While both methods have been shown to yield similar results (e.g., Mirman & Graziano, 2013), each has its advantages and disadvantages as it relates to the current study. Lesion subtraction analysis contrasts voxels that are lesioned in a group of participants that exhibit an impairment of interest with those that do not exhibit said impairment (here, the not improved vs. the improved group, respectively). To establish statistical significance in terms of lesion location differences, this method uses a principled difference threshold to determine a meaningful difference between more frequently lesioned regions in one group vs. the other, which in turn accounts for the fact that some regions are more vulnerable for cerebrovascular reasons rather than implying a functional link. However, because this method involves defining groups of participants based on a difference in their behavioral profile, it does not allow for controlling for other factors that may impact that behavioral difference, such as lesion volume. This is particularly relevant to the current study because here we expect lesion volume to play a significant role. In order to identify regions that are associated with poor long-term outcomes while controlling for differences in total lesion volume, we

conducted a VLSM on the residuals of percent change in WAB-R AQ after controlling for lesion volume. However, this approach has its limitations in that it involves conducting multiple t tests at each voxel comparing the size of the behavioral effect (or impairment of interest) for individuals with and without a lesion in that voxel. Because this requires repeated t tests, it is standard practice to correct for multiple comparisons, which can often be too conservative when the study involves a limited number of participants. Thus, we implemented the lesion subtraction analysis because it offers a measure of statistical significance that is acceptable given the sample size of the current study, but we report uncorrected VLSM results to address the concern of group-based differences in total lesion volume that could also explain overall differences in long-term outcomes. Details pertaining to each analytic approach are described below.

We first assessed whether the lesion location critically differed between participants who improved vs. those who did not using a lesion subtraction analysis (Karnath et al., 2002). Participants were assigned to one of two groups depending on degree of WAB-R AQ change from initial to follow-up testing using the criterion of a greater than 5-point increase in AQ (Elman & Bernstein-Ellis, 1999; Holland et al., 2017; Maher et al., 2006) to define clinically meaningful improvement in language abilities (see Results section for details). We computed a lesion overlap map for those whose WAB-R AQ increased by five or more points (“improved” group, $n = 12$) and one for those whose WAB-R AQ remained the same or decreased (“not improved” group, $n = 6$). We then subtracted the lesion overlap map of the “improved” group from that of the “not improved” group, and the resultant map was thresholded at 50% to identify voxels with the largest group differences. Specifically, we considered only voxels where at least 50% more participants in the “not improved” group had a lesion than those in the “improved” group (e.g., if six participants have a lesion to a given voxel, then the 50% threshold would include only those voxels with two participants in the “improved” and four in the “not improved” group). This threshold also constrains the voxels included in the analysis to those in which lesions are present in at least 3 out of the 6 participants who did not improve in the chronic phase. Lesion subtraction analysis was performed in the software package MRICron (www.mccauslandcenter.sc.edu/mricron/mricron). Gray matter regions were identified using the automated anatomical labeling (AAL) atlas (Tzourio-Mazoyer et al., 2002) and white matter pathways were identified using the Catani atlas (<https://www.natbrainlab.co.uk/atlas-maps>).

Given that total lesion volume may account for group differences in lesion location, as identified in the lesion subtraction analysis, we additionally performed a VLSM analysis with percent change in WAB-R AQ residuals after regressing out variance due to total lesion volume. Voxels in which fewer than 20% of participants had a lesion ($n = 4$) were excluded from the analysis. We report uncorrected VLSM results since (1) the sample size is relatively small for this type of analysis, and (2) the analysis was meant to provide evidence that the regions identified in the lesion subtraction analysis could not fully be accounted for by variability in total lesion volume (especially between the two groups of participants).

Results

Initial WAB-R assessments were performed from 5–133 months post-onset (MPO) ($M = 40$), and follow-up tests from 21–94 months after initial testing ($M = 61$). The difference in WAB-R AQ from initial to follow-up testing ranged from -7.2 to 24.7 points. AQ scores for 12 participants increased by 7 or more points; for the remaining 6, AQ scores remained the same or decreased. This bifurcation of the change scores enabled us to subdivide the sample into “improved” ($n = 12$) and “not improved” ($n = 6$) groups (Table 1) for the lesion subtraction analysis. Importantly, a 7-point increase on the WAB-R exceeds the 5-point criterion for clinically meaningful improvement in prior clinical studies (Elman & Bernstein-Ellis, 1999; Holland et al., 2017; Maher et al., 2006), and one participant achieved what is considered full recovery (i.e., AQ score ≥ 93.8).

Regression results

Multiple regression assessing previously identified predictors of aphasia recovery

—Inspection of the P-P plot revealed that inclusion of the factor sex violated the assumption of normally distributed residuals, and this factor was therefore removed from the model. The final regression model with four predictor variables explained 51.9% (unadjusted R^2) of the variance, $F(4, 13) = 3.51$, $p = .037$, $R^2_{\text{Adjusted}} = .37$. As shown in Table 2, lesion volume was the only significant predictor of proportion change in AQ. The negative regression weight indicates that smaller lesions yield larger language gains in the long term, after controlling for other predictors included in the model. No other predictor significantly contributed to the model (see Figure 1).

Multiple regression exploring aphasia recovery as a function of psychosocial and treatment-related factors

—The final model exploring the relationship between psychosocial and treatment-related factors explained 56.1% (unadjusted R^2) of the variance, $F(3, 14) = 5.96$, $p = .008$, $R^2_{\text{Adjusted}} = .47$, which retained three factors: ALA scores on the aphasia and participation domains as well as amount of SLT received. As shown in Table 3, ALA aphasia and participation domain scores significantly related to proportion change in AQ, whereas the amount of SLT trended towards significance. The positive regression weight for ALA participation domain indicates that greater satisfaction with life participation was related to better long-term outcomes. Similarly, the positive – albeit not statistically significant – regression weight for amount of SLT points to a positive benefit of receiving several courses of SLT. However, the negative regression weight for the ALA aphasia domain suggests that those who perceived their language impairment as more severe achieved larger language gains (see Figure 2).

Multiple regression assessing added predictive value of psychosocial and treatment-related factors on long-term outcomes

—The stepwise regression assessing whether the predictors identified as impacting outcomes in the psychosocial and treatment-related regression model add predictive value above and beyond that identified in the model of previously known predictors of recovery (i.e., lesion volume) revealed that adding the ALA aphasia and participation scores along with the amount of SLT received significantly improved model fit, $F(3, 13) = 3.66$, $p = .04$. The final model explained

68.2% (unadjusted R^2) of the variance, $F(4, 13) = 6.96$, $p = .003$, $R^2_{\text{Adjusted}} = .58$, which retained three factors: lesion volume as well as ALA scores on the aphasia and participation domains. Amount of SLT, however, was not a significant predictor in the model. Consistent with the previous regression models, the negative regression weights for lesion volume and ALA aphasia domain scores indicate that smaller lesions and greater, self-perceived aphasia severity relate to better outcomes. By contrast, the positive regression weight for ALA participation domain suggests that increased satisfaction with life participation is associated with greater long-term language gains. While the finding that perceived severity of impairment related to worse outcomes may be somewhat counterintuitive, it is not entirely unexpected for a number of reasons, as explained in more detail in the Discussion.

To ensure that the multiple regression findings were not driven by the two participants that were first assessed at the cusp of the subacute/chronic threshold (i.e., 5 MPO), we ran the final stepwise regression model excluding these participants. The results were partially consistent with those involving all participants, as lesion volume remained a significant predictor of long-term language gains, $p < .05$, but psychosocial and treatment-related factors were not, p 's $> .27$. We discuss potential reasons for this in the Discussion.

Lesion location results

As illustrated in the lesion overlap map depicted in Figure 3a, participants in the “not improved” group all have large lesions that encompass the core perisylvian language network, whereas lesion location for those in the “improved” group is more variable, i.e. maximal overlap across the group occurred in 10 out of 12 participants. The lesion subtraction analysis points to several gray and white matter regions that may be critical to predicting aphasia prognosis (Figure 3b). Compared to the “improved” group, participants in the “not improved” group were more likely to have lesions encompassing the temporoparietal and frontal cortices. Temporoparietal regions that survived the threshold included the left inferior and superior parietal gyri; superior, middle, and inferior temporal gyri (STG, MTG, and ITG respectively) as well as the superior temporal pole; and the angular gyrus as well as Heschl's gyrus. Frontal regions included the middle frontal gyrus; Broca's area (pars triangularis, opercularis, and orbitalis) in the inferior frontal gyrus; the pre- and post-central gyrus; and insular cortex. The subtraction map also revealed that individuals who did not improve vs. those who did were more likely to have damage to subcortical white matter structures connecting these regions, including the left posterior and anterior segment of the arcuate fasciculus (AF), internal capsule, inferior fronto-occipital fasciculus (IFOF), inferior longitudinal fasciculus (ILF), and the uncinate fasciculus (UF). Other white matter tracts that survived the threshold included the cingulum, corticospinal tract, and corpus callosum.

As shown in Figure 3c, the uncorrected VLSM results are consistent with the lesion subtraction analysis, indicating that the difference in total lesion volume – as identified in the regression analyses – do not fully account for the differences in lesion location based on the lesion subtraction analysis. Thus, even after accounting for variance in total lesion volume, there appear to be crucial regions that when damaged may impact the potential for long-term recovery.

Discussion

This study explored the potential influence of patient-, stroke-, and treatment-related factors on long-term recovery from chronic aphasia. We conducted this study following the incidental finding that some individual's language abilities substantially improve within the chronic phase. Lesion site may also impact the potential for aphasia recovery in the long term, with lesions localised to the frontal and temporoparietal cortices along with the underlying white matter pathways connecting these regions yielding poorer long-term outcomes. Additionally, satisfaction with life participation and self-perceived aphasia severity may represent patient-related variables significant to aphasia recovery, although for reasons discussed below, it is more difficult to make strong causal inferences about their influence. Lastly, we provide evidence potentially suggestive of the positive impact that receiving multiple courses of SLT has on long-term outcomes, though this factor only trended towards significance and does not improve model fit when lesion volume and other psychosocial variables are taken into consideration. Thus, the results from this study not only challenge the widely accepted belief that aphasia recovery is static in the chronic stage, but they also shed light on factors that may promote aphasia recovery in both acute and chronic stages.

Our findings complement and extend those reported in recent work (Basilakos et al., 2019; Holland et al., 2017; Hope et al., 2017; Johnson et al., 2019), as we find that language gains are possible in the chronic phase and may be tied to a number of factors present both immediately following and long after stroke onset. In particular, we demonstrate that lesion volume plays a significant role in long-term aphasia outcomes, which is consistent with studies of aphasia recovery over time frames closer to stroke onset (Maas et al., 2012; Plowman et al., 2012; Szaflarski et al., 2013). While these data suggest that smaller lesions allow for functionally beneficial reorganization of the language network, it is important to note that lesion volume was *not* associated with changes in aphasia severity in the two studies to investigate factors tied to changes in chronic aphasia severity (Basilakos et al., 2019; Hope et al., 2017). The discrepancy between their findings and ours is likely due to differences in the analytic approach. Hope et al. (2017) compared lesion volumes for individuals who improved vs. declined using a different measure of aphasia severity (i.e., the Comprehensive Aphasia Test [CAT]; Swinburn et al., 2004) than that used here, and it is not clear whether the threshold for assigning participants to each group is clinically meaningful. Using a regression analysis, Basilakos et al. (2019) found that lesion volume did *not* predict changes in aphasia severity with the inclusion of a measure of white matter hyperintensity severity (i.e., leukoaraiosis as measured via Fazekas scale scores) as a predictor, suggesting that lesion volume may not account for additional variance that can be explained by leukoaraiosis severity. Because leukoaraiosis is also associated with regional gray matter loss (Lambert et al., 2016), it could suggest that the relationship between lesion volume and aphasia outcomes observed here is due in part to pathological white matter hyperintensities. Future work is needed to disentangle potential shared and separable contributions of gray and white matter injury on chronic post-stroke aphasia recovery.

Here, we provide some evidence to suggest that the location of damage within the perisylvian language network is a likely predictor of long-term aphasia recovery. In

the current study, damage to language structures within and between the frontal and temporoparietal cortices was associated with poor long-term outcomes. Temporoparietal regions identified here have been associated with worse language recovery from acute/subacute to chronic stages, including the posterior STG (Hanlon et al., 1999; Plowman et al., 2012; Sul et al., 2016), posterior segment of the AF (equivalent to the superior longitudinal fasciculus [SLF] in the John Hopkins University – Montreal Neurological Institute [JHU-MNI] atlas, cmrm.med.jhmi.edu; Hillis et al., 2018), and the ILF (Ramsey et al., 2017). Damage to frontal regions, such as Broca's area, the insular cortex, and underlying white matter, has also been tied to poor spontaneous recovery from the subacute to chronic stage (Sul et al., 2016). However, damage to these frontal structures appear to specifically affect fluent speech production (Naeser et al., 1989; Sul et al., 2019), rather than overall aphasia severity, and frontal damage alone is not sufficient to account for changes in language abilities in the long term (Naeser et al., 1998; Ramsey et al., 2017). In the current study, we provide some evidence to suggest that the difference in lesion location for people who were more vs. less likely to exhibit long-term language gains was not driven by differences in lesion volume alone, as the VLSM analysis controlling for lesion volume yielded comparable results. Taken together, the results suggest that damage to specific posterior and anterior regions crucial for language along with the connections between them impedes long-term aphasia recovery (see also Gleichgerrcht et al., 2016). This may be because lesions that encompass more of these language-relevant centers interfere with compensatory reorganization of language processing, which may be better facilitated when there is relative sparing of language-relevant left hemisphere regions.

Regarding patient-related factors, our findings provide some support for the influence of psychosocial – but not demographic – variables, but the results should be interpreted with caution. First, although the positive relationship between scores on the ALA participation domain and long-term improvements suggests that greater satisfaction with participation in life situations related to better recovery, the negative relationship between the ALA aphasia domain and recovery is somewhat counterintuitive. It is possible that there is an inherent relationship between these two variables, such that one who perceives improvement in their language abilities also perceives greater satisfaction in life participation (i.e., increased participation in life activities due to feeling less concerned about language impairment severity). Indeed, inspection of this relationship reveals that perceived language abilities is positively related to perceived satisfaction with life participation (Spearman's $Rho = .56$, $p = .02$), suggesting that these predictors should impact recovery in a similar fashion. However, because multiple regression analysis estimates the effect of a predictor while holding all other predictor variables constant, it suggests that these two factors may be dissociable given the direction of their regression weights. While speculative, this finding may reflect differences in baseline aphasia severity, and in turn potential for improvement over time, and/or personal attitudes regarding current level of impairment. Nonetheless, that satisfaction with life participation dissociated from perceived level of impairment is consistent with prior work, which also indicates that participation is a better correlate of subjective quality of life when compared to impairment severity (e.g., Eadie et al., 2006; see also Simmons-Mackie et al., 2014). Thus, the positive relationship between the ALA participation domain and long-term outcomes likely reflects a positive impact of quality of

life on aphasia recovery in the long term. Second, because we did not obtain these data at initial assessment, we cannot address whether the relationship between these psychosocial variables and aphasia recovery reflects attitudes that were stable across the chronic period or specific to the follow-up assessment. However, consistent with prior work, we did not find any relationship between long-term recovery and sex (Hope et al., 2017), education (Basilakos et al., 2019; Johnson et al., 2019), or age (Holland et al., 2017; Hope et al., 2017). Thus, although satisfaction with life participation, but not self-perceived language impairment severity, may mediate long-term aphasia recovery, future research is needed to better understand how stable personality factors vs. situational factors (presumably influenced by environment) contribute to long-term aphasia outcomes.

Lastly, our results provide evidence suggestive of SLT influencing long-term outcomes, but cannot definitively speak to the impact of other treatment-related variables. That is, the amount of SLT (in terms of number of courses of treatment) received was positively related to long-term language gains, which is consistent with a previous study assessing aphasia recovery in the chronic stage (Johnson et al., 2019); however, in the current study, this factor only trended toward statistical significance. This may be because we used an approximate measure of amount of SLT, rather than the more objective measure of SLT (i.e., number of treatment hours) used in Johnson et al. (2019). Responses on many of the other treatment-related variables queried either had no variability or were unreliable. For instance, many participants ($n = 8$) were unable to reliably self-report whether or not they received tPA following stroke onset and almost all participants ($n = 15$) reported attending rehabilitation, precluding inferences regarding their effect on long-term outcomes. However, because thrombolytic treatments decrease the extent of damaged tissue (Silvennoinen et al., 2008), patients who received tPA or mechanical clot retrieval likely have smaller lesions, and in turn, better long-term outcomes. It will be important for future research to prospectively investigate treatment-related factors associated with long-term outcomes or retrospectively obtain hospital records to determine whether specific interventions were administered.

Of note, adding psychosocial and treatment-related factors to significant predictors from amongst the previously established factors of long-term recovery (i.e., lesion volume) significantly improved model fit in the full sample of participants, indicating that these factors add predictive prognostic value of long-term language gains following stroke. To our knowledge, this is the first study to demonstrate that environmental factors, including social support and access to support groups (or potentially other forms of treatment), impact long-term trajectories and therefore could potentially compensate for the impact of the extent of neurological damage to the language network. This result, however, must be interpreted with caution for a few reasons. First, as mentioned above, the self-report nature of these measures and the fact that they were collected at the follow-up (but not initial assessment) makes it difficult to discern whether self-perceptions reflect continuity (either in personality or social support systems) or situational factors that may influence responses on questions of this type. Second, the analysis excluding the two participants on the cusp of the “chronic” stage confirmed the predictive value of lesion volume, but did not reveal any relationship between psychosocial/treatment-related variables and the potential for long-term recovery, which may call into question the predictive value of the psychosocial/treatment-related factors (see e.g. Steyerberg, 2009). However it’s possible that psychosocial and treatment-related

factors, at least as measured here, evolve over time since post-stroke onset, meaning that patient-reported outcomes may change over time depending on how much time has elapsed since stroke onset and one's satisfaction with their (self-perceived) degree of recovery at that time point. Nonetheless, future research with larger sample sizes sampling from a range of participants at various time points post-stroke onset is needed to confirm this hypothesis.

Taken together, the findings from this study paint a more promising picture for individuals with chronic aphasia and offer new insight into factors that may facilitate long-term aphasia recovery. However, because this study was conducted retrospectively on a small group of people with aphasia who represent a self-selected sample of "convenience," it will be important for future studies to replicate these findings using a prospective approach in a larger sample. It is also worth noting that although the lesion subtraction results reported here are largely consistent with previous literature (Hope et al., 2013; Rosso et al., 2015) and are also in line with the uncorrected VLSM results, the findings should be interpreted with caution due to the small number of participants included as a whole, which can impact the generalizability of either result. Our implementation of stringent thresholds likely helps to circumvent the issues associated with small sample sizes; however, using this approach to identify white matter structures may be less reliable than using a tractography-based approach. High-resolution diffusion tensor imaging using tractography-based approaches to explore how damage to different white matter pathways influences long-term aphasia outcomes is needed to confirm our findings.

Conclusion

We believe this study has direct clinical implications for both the acute and chronic phases of stroke-aphasia care. Our finding that lesion size was a major predictor of recovery in the chronic phase underscores the need for acute stroke care to focus on tissue preservation, whether through pharmacologic/procedural interventions that minimise the ischemic burden or by increasing community awareness and education to enable early detection of and response to stroke symptoms. Our broader finding that some people with aphasia continue to show language improvement years after stroke onset suggests that providing long-term therapeutic interventions could be helpful. This stands in sharp contrast to current clinical practice, wherein insurers often provide coverage for treatments like SLT for only a brief time period after stroke. Lastly, it may be helpful to make a range of resources available for people with chronic aphasia that enhance life participation in those with persistent difficulties – especially in light of the fact that continued improvement may be possible for many of them.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

We thank Myrna F. Schwartz for consultation on design, analysis, and statistical approach as well as feedback on earlier drafts of the manuscript. We also thank Claire Healy for assisting with collecting follow-up language assessment data and follow-up self-report data as well as H. Branch Coslett for overseeing the lesion tracings.

This work was supported by the NIH-NIDCD under Grant R01-DC012780 to RHH, and NIH-NICHHD under Grant T32-HD071844 to DYH.

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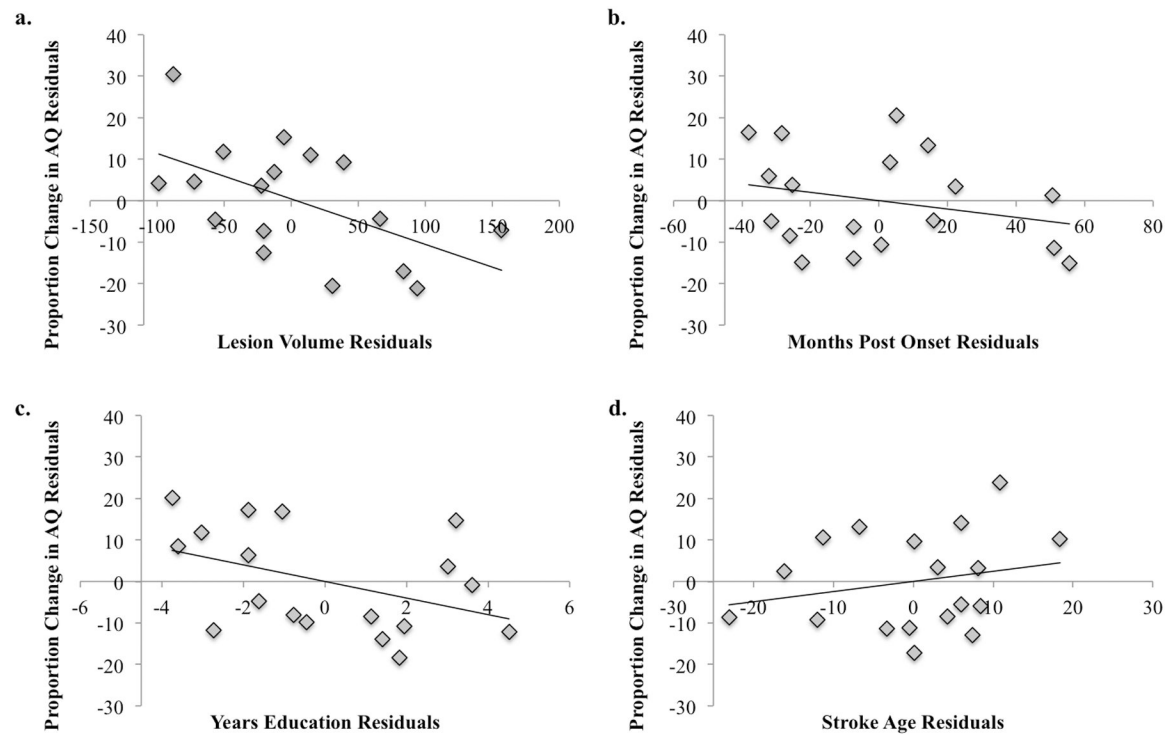


Figure 1.

Partial correlation plots for the multiple regression analysis assessing previously identified prognostic indicators of aphasia recovery. Proportion change in AQ residuals are plotted against (a) lesion volume residuals, (b) months post onset residuals, (c) years education residuals, and (d) stroke age residuals.

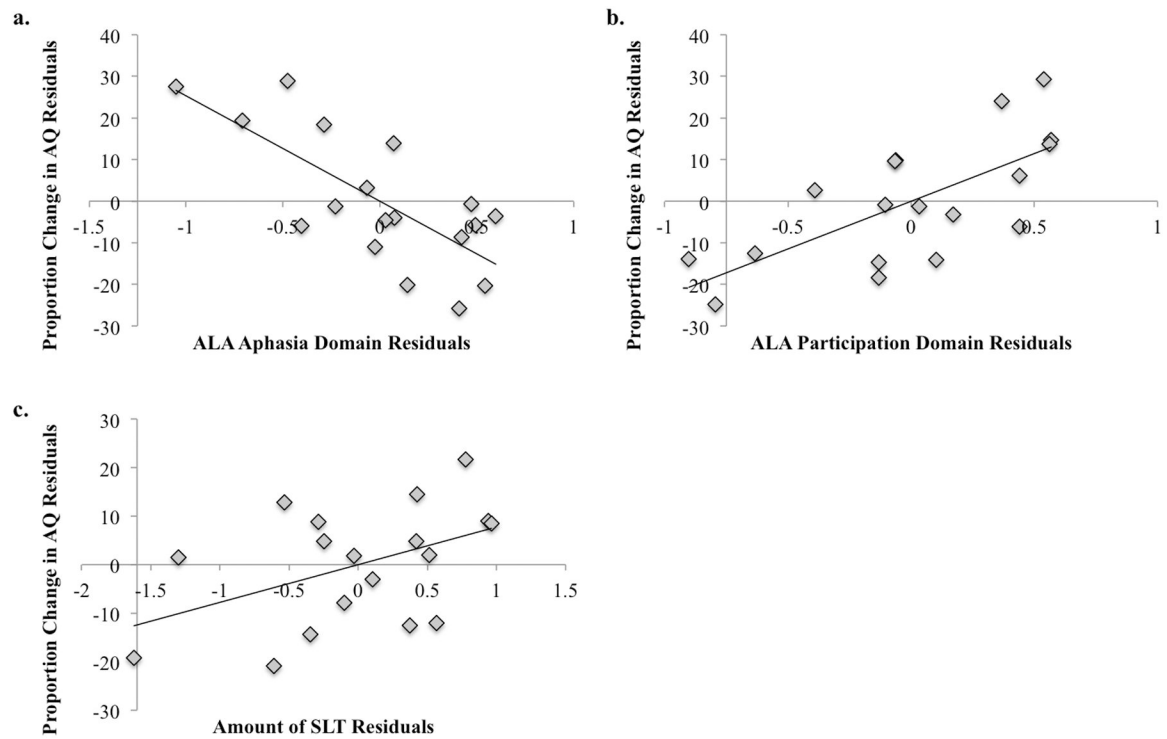


Figure 2.

Partial correlation plots for the multiple regression analysis exploring psychosocial and treatment-related factors as indices of aphasia recovery. Proportion change in AQ residuals are plotted against (a) ALA aphasia domain residuals, (b) ALA participation domain residuals, and (c) amount of SLT residuals.

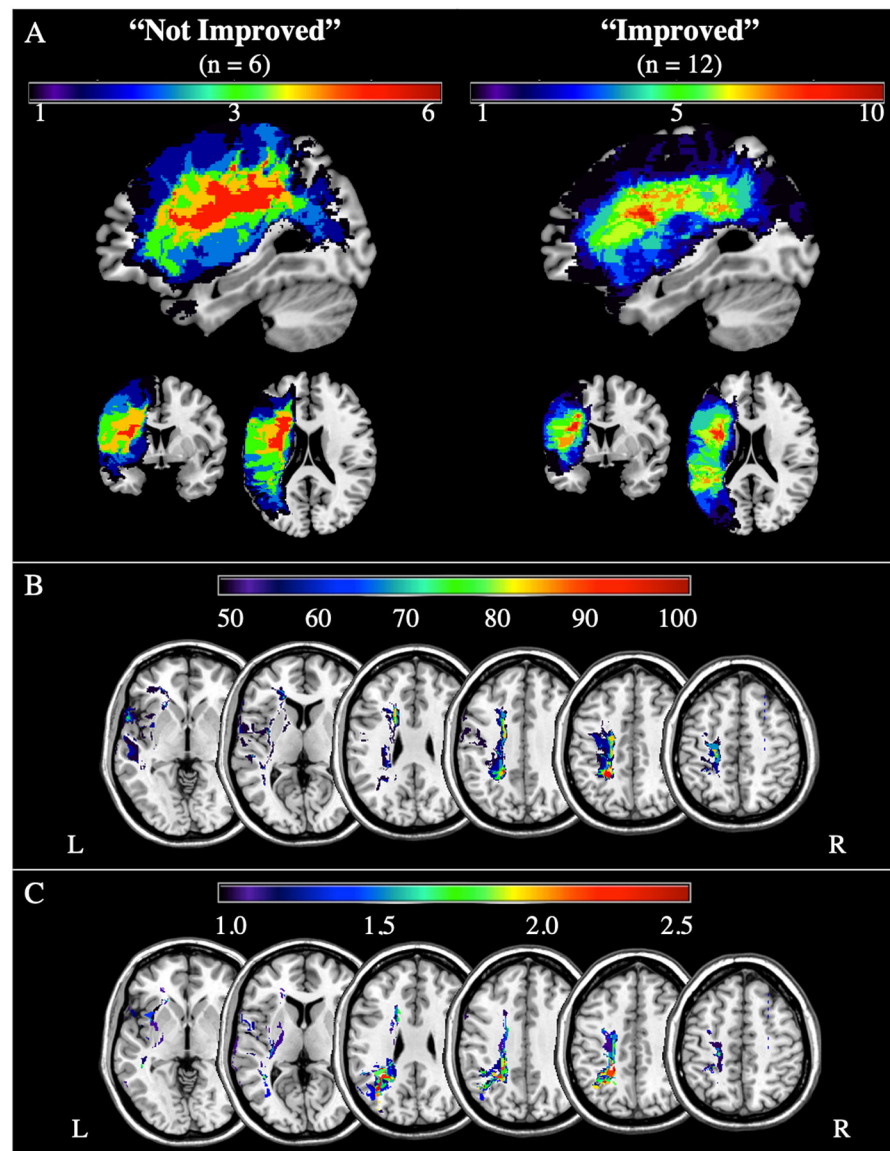


Figure 3. Lesion location comparisons for the "improved" and "not improved" groups. Panel A: Lesion overlap maps for the individuals in the "not improved" (left) and "improved" (right) groups. Panel B: Lesion subtraction map depicting regions where the "not improved" group had at least 50% more lesions than the "improved" group. Panel C: The uncorrected VLSM results thresholding to include only lesions present in at least 20% of participants (n = 4), where the color scale represents uncorrected t values ranging from 1 to 2.5. In Panel B and Panel C, slices correspond to -3, 7, 29, 36, 43, and 50mm in Montreal Neurological Institute (MNI) coordinates.

Table 1

Demographic, Clinical, and Neuroanatomical Information for the Participants

Subject ID	Group	Sex	Yrs Ed	Age		Months Post-onset		WAB-R Classification		AQ		Les Vol (cc)
				Onset	Initial	Follow-up	Initial	Follow-up	Initial	Follow-up	% Change	
S01	NI	M	18	49	51	58	19	109	B	B	55.6	219.93
S02	NI	M	19	54	61	68	80	173	B	B	48.4	356.21
S03	NI	M	12	57	68	76	133	227	B	B	69.5	197.30
S04	NI	F	12	43	45	49	28	74	A	A	84.2	140.55
S05	NI	M	15	55	57	62	28	88	A	A	88.2	195.29
S06	NI	M	19	63	64	70	13	89	A	A	91.2	84.92
S07	I	F	16	39	46	54	80	173	C	A	90	151.32
S08	I	M	12	69	77	79	99	123	B	B	41.6	147.65
S09	I	F	13	40	47	50	81	117	A	A	81.5	85.21
S10	I	F	18	31	31	36	6	56	C	C	86.3	63.92
S11	I	F	12	40	45	49	65	114	B	B	43.7	153.99
S12	I	M	14	45	47	51	23	63	A	A	83.2	55.69
S13	I	M	19	62	62	65	6	47	A	Recov.	94.6	61.03
S14	I	M	14	47	49	57	20	111	B	B	77.8	172.21
S15	I	F	12	29	30	34	11	55	B	TMA	75.8	147.24
S16	I	M	12	52	53	58	14	78	C	A	85.6	57.64
S17	I	M	12	48	48	50	5	26	B	B	77	89.60
S18	I	M	21	72	72	79	5	89	C	A	80.8	8.01
Mean			15	49.6	52.9	58.1	39.8	100.7			75.7	132.6
SD			3.2	12	12.7	13.1	39.2	50.4			16.9	81.1
Range			12-21	29-72	30-77	34-79	5-133	26-227			41.6-94.6	8-356.2

Note. Participants are arranged in order of WAB-R AQ difference scores. NI = "not improved" group; I = "improved" group; F = female; M = male; Yrs Ed = years education; A = Anomic; B = Broca's Aphasia; C = Conduction Aphasia; TMA = Transcortical Motor Aphasia; Recov. = Recovered; WAB-R = Western Aphasia Battery-Revised; AQ = Aphasia Quotient; Les Vol (cc) = lesion volume (in cubic centimeters); SD = standard deviation.

Table 2

Results from the Multiple Linear Regression Assessing Previously Identified Prognostic Indicators of Aphasia Recovery

	<i>b</i>	<i>SE b</i>	β
Constant	48.71	18.61	
Lesion Volume **	-0.11	0.05	-0.53
MPO	-0.10	0.10	-0.24
Stroke Age	0.24	0.30	0.18
Years Ed	-2.00	1.19	-0.39

Note. The dependent variable was proportion change in WAB-R AQ. $R^2 = .519$, $R^2_{\text{Adjusted}} = .37$. *b* = unstandardized regression weights; *SE* = standard error; β = standardized regression weights; MPO = months post onset (at the initial assessment).

*
 $p < .10$,

**
 $p < .05$,

 $p < .01$

Table 3

Results from the Multiple Linear Regression Exploring Psychosocial and Treatment-related Factors as Indices of Aphasia Recovery

	<i>b</i>	<i>SE b</i>	β
Constant	-17.73	18.75	
ALA Aphasia Domain ***	-25.34	6.26	-0.97
ALA Participation Domain **	22.94	6.34	0.94
Amount of SLT *	7.77	4.03	0.40

Note. The dependent variable was proportion change in WAB-R AQ. $R^2 = .561$, $R^2_{\text{Adjusted}} = .47$. *b* = unstandardized regression weights; *SE* = standard error; β = standardized regression weights; ALA = Assessment for Living with Aphasia; SLT = speech-language therapy.

*
 $p < .10$,

**
 $p < .05$,

 $p < .01$

Table 4

Results from the Multiple Linear Regression Evaluating Added Predictive Value of Psychosocial and Treatment-related Factors When Compared to Lesion Volume as Indices of Aphasia Recovery

	<i>b</i>	<i>SE b</i>	β
Constant	-17.73	18.75	
Lesion Volume **	-0.08	0.04	-0.40
ALA Aphasia Domain ***	-19.99	6.03	-0.76
ALA Participation Domain **	15.65	6.48	0.64
Amount of SLT	5.26	3.73	0.27

Note. The dependent variable was proportion change in WAB-R AQ. $R^2 = .682$, $R^2_{\text{Adjusted}} = .58$. *b* = unstandardized regression weights; *SE* = standard error; β = standardized regression weights; ALA = Assessment for Living with Aphasia; SLT = speech-language therapy.

*
 $p < .10$,

**
 $p < .05$,

 $p < .01$

Color legend

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claim	#a8c8ff	
confounders	#cccccc	
imaging_details	#d9b3e6	
limitation	#ffb3b3	
method	#fff4a3	
region_definition	#d9b3e6	
sample	#a8e6a3	
statistics	#ffcc99	